	A	B	C	D	E	F	G	H	I
1	An observer wishes to estiamte the population proportion of smoker by using sampling technique								
2	with error not more than 8%. If previous study shows that proportion of smoker was 30%.								
3	Compute the appropriate sample size								
4		(i) If he wishes to be 95% confident.							
5		(ii) If he wishes to 99% confident.							
6		(iii) If he wishes to be almost certain and information about pop.							
7		propotion is not given.							
8									
9									
10	Solution:- We have,								
11		Max. permissable error (E) =			0.08				
12			Pop. Prop.(P) =		0.3				
13			Q =		0.7				
14		For (i) Here, C.I.(1-α) =			0.95				
15				α =	0.05				
16				Zα =	1.959964	=NORMSINV(1-E15/2)			
17	Now, Required sample size (n) =				126.0479	=(E16/E11)^2*E12*E13			
18		Hence, Required sample size is 127.							
19		For (ii) Here, C.I.(1-α) =			0.99				
20				α =	0.01				
21				Zα =	2.575829	=NORMSINV(1-E20/2)			
22	Now, Required sample size (n) =				217.7075	=(E21/E11)^2*E12*E13			
23		Hence, Required sample size is 12			218				
24	For (iii) Here, For almost certainly lie,								
25				Zα =	3				
26	If value of p is not given , We use P =				0.5				
27				Q =	0.5				
28	Now, Required Sample Size (n) =				351.5625	=(E25/E11)^2*E26*E27			
29									
30		Hence, Required sample size is 352.							
31									
32					Name:-Hari Bashyal				